

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460236.7466 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.0064
Transit depth (Rp/Rs)^2	2.89 +/- 0.22 [%]
Orbital Inclination (inc)	88.06 +/- 1.14
Airmass coefficient 1 (a1)	0.751 +/- 0.014
Airmass coefficient 2 (a2)	0.238 +/- 0.041
Scatter in the residuals of the lightcurve fit is	1.79 %
Best Comparison Star	#2 - [78, 422]
Optimal Aperture	4.13
Optimal Annulus	6.18
Transit Duration (day)	0.0777 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.0064 vs 0.1618 (deviation 0.87σ)
Duration fit vs published	0.0777 d vs 0.0721 d (deviation 0.54σ)
Mid-time O–C	27.2 min
Depth SNR	18.0
Residual scatter	1.79 %

Light curve

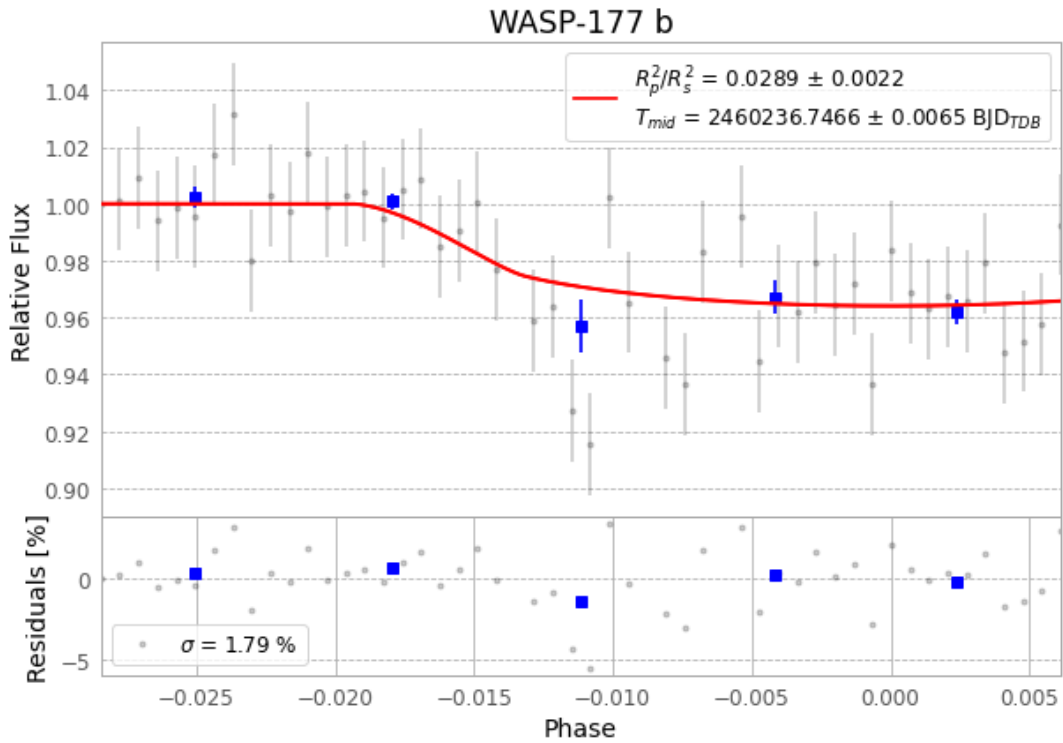


Figure 1 — FinalLightCurve_WASP-177 b_2023-10-18.png (SHA-256 6145ba6d27bd2ee9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 284a11c23b8bbdb1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

52 FITS frames (34 MB), WASP-177231019034215.FITS → WASP-177231019061515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-231019.log (8258cadd0858...), FinalLightCurve_WASP-177 b_2023-10-18.png (6145ba6d27bd...), FinalLightCurve_WASP-177 b_2023-10-18.csv (42d4730b881c...)

Compute resources

Wall time 302.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:54Z.